HARSHIT MISHRA
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HARSHIT MISHRA

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RESEARCH AND PUBLICATIONS

- Current Work: Node Classification in Heterophilic Text Attributed Graphs. Use Multi Head Attention, GraphSAGE and GCN (GNN) models to deliver state-of-the-art accuracy.
- Harshit Mishra and Sucheta Soundarajan. Addressing Bias in Information Retrieval. Part of the book series: SpringerBriefs in Intelligent Systems
- Harshit Mishra and Sucheta Soundarajan. A Case Study of Balanced Query Recommendation on Wikipedia. Accepted at FAccTRec 2025 workshop at recsys 2025
- Harshit Mishra and Sucheta Soundarajan. BalancedQR: A framework for balanced query recommendation. In Proceedings at ECML/PKDD 2023
- Harshit Mishra and Sucheta Soundarajan. Keyword Recommendation for Fair Search. In Proceedings at Bias 2022 workshop at 44th European Conference on Information Retrieval, ECIR '22. Springer, 2022
- Harshit Mishra, Sucheta Soundarajan. Keyword Recommendation for Fair Search. In the IJCAI 2021 Workshop on AI for Social Good.
- Harshit Mishra. Reducing Word Embedding Bias. At an AI for Social Good, Harvard, CRCS Workshop 2020.

EMPLOYMENT

Teaching/ Research Assistant — **Syracuse University** — **August 2024 - Present**

- Researching node classification models on heterophilic network systems.
- Development of a personalized balanced query recommendation method.
- Evaluation of fairness metrics for information retrieval.
- Teaching Assistant: Design & Analysis of Algorithms, Applied Natural Language Processing, Fundamentals Computing and Programming

Software Engineer — **MicroStrategy** — **June 2021 – August 2024**

- Built an ML-LLM product to analyze time series data and provide interesting observations to end users.
- Developed ML and deep learning models to provide anomaly detection and trend insights based on user data.

Research Intern — **Syracuse University** — **June 2020-Jan 2021**

- Developed a query reformulation method to reduce bias present in search engine results.
- Presented a novel method to reduce bias in word embeddings using variational auto encoders.

EDUCATION

Syracuse, NY — **Syracuse University** — **August 2024 - Present**

- PhD Student in Computer and Information Science.
- Coursework: Reinforcement learning, Graph Theory, Artificial Neural Networks, Convex Optimization, Algorithms.

Syracuse, NY — **Syracuse University** — **August 2018 – May 2020**

- M.S. in Computer and Information Science, May 2020.

SELECTED PROJECTS

- **Talk Poetry** (2026). Used codex and GPT-luna, GPT-40-mini-tts models to create a voice-based poetry app. Intended use of the app is to alleviate loneliness by discussing poetry with a LM.
- **Multi-dimensional BQR** (2024). Used reinforcement learning method, UCB1 as a multi arm bandit approach to learn user behavior and re-rank BQR queries.
- **Debiasing Word Embeddings** (2020). Removed gender bias by 3% in word embeddings using variational autoencoders and adaptive sampling. Python, ML, PyTorch, Deep Learning, Variational Autoencoders.
- **Twitter EchoChambers** (2018). Validated presence of echo chambers on Twitter by showing similar and dissimilar networks of twitter followers for democratic and republican leaders.

DISTINCTIONS AND AWARDS

- **Veracode:** Earned a Veracode Level 2 certification as part of of my quarterly goals at MicroStrategy.
- **KCNA:** Earned a KCNA: Kubernetes and Cloud Native Associate certification as part of my quarterly goals at MicroStrategy.
- **Hackathon:** Co-winner a company-wide hackathon.

LANGUAGES AND TECHNOLOGIES

- Python, PyTorch, Transformers, Scikit-learn
- RAG, PEFT, LoRA,
- GCN, GAT, GraphSAGE, Neural Nets
- LLMs, NLP, Information Retrieval, Network Science, Graph Machine Learning, Reinforcement Learning